Gene Pulser® Electroprotocol

Cell Type Other Cell Types

Species Used Chicken, HD11, macrophage

Molecules Electroporated DNA: linearized DNA used for stable transfections.

Before the Pulse

Cell Growth Medium DMEM, 8% Fetal Calf Serum (FCS), 2% Chicken Serum (GIBCO/BRL, Sigma)

Growth Phase at Harvest 50 to 70% confluency

Pre-pulse Incubation 4° C, 10 min.

Wash Solution Wash two times in electroporation buffer

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature Room temperature

Electroporation Medium Phosphate Buffered Saline

Cell Density 5 x 10 (5) cells/pulse, stable transfection

Volume of Cells 0.5 ml

DNA Concentration 10 µg / pulse

DNA Resuspension Buffer Not given; pulse volume: 0.8 ml

Volume of DNA Not given; pulse volume: 0.8 ml

Cuvette Gap 0.4 cm

Voltage 0.270 kV

Field Strength 0.675 kV/cm

Capacitor 960 µF

Resistor (Pulse Controller) Ω none

Time Constant 28 msec

After the Pulse

Outgrowth Medium DMEM, 8% Fetal Calf Serum (FCS), 2% Chicken Serum

Outgrowth Temperature 37 °C

Length of Incubation 48 to 72 hrs.

Selection Method or Assay Used G418 (stable transfections)

Electroporation Efficiency Not given

Per Cent Survival about 50%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
PBS: 1x = 8g NaCl, 0.2g KCl, 0.2g KH_2PO_4, 1.15g Na_2HPO_4

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Survey Number 186